3.1.10 Equilibrium constant K_p for homogeneous systems (A-level only)

The further study of equilibria considers how the mathematical expression for the equilibrium constant K_p enables us to calculate how an equilibrium yield will be influenced by the partial pressures of reactants and products. This has important consequences for many industrial processes.

Content	Opportunities for skills development
The equilibrium constant K_p is deduced from the equation for a reversible reaction occurring in the gas phase. K_p is the equilibrium constant calculated from partial pressures for a system at constant temperature. **Students should be able to:** • derive partial pressure from mole fraction and total pressure • construct an expression for K_p for a homogeneous system in equilibrium • perform calculations involving K_p • predict the qualitative effects of changes in temperature and pressure on the position of equilibrium • predict the qualitative effects of changes in temperature on the value of K_p • understand that, whilst a catalyst can affect the rate of attainment of an equilibrium, it does not affect the value of the equilibrium constant.	**MS 1.1** Students report calculations to an appropriate number of significant figures, given raw data quoted to varying numbers of significant figures. Students understand that calculated results can only be reported to the limits of the least accurate measurement. **MS 2.2 and 2.3** Students calculate the partial pressures of reactants and products at equilibrium. Students calculate the value of an equilibrium constant K_p